

Excise: One-Shot Capability Extraction from Open Language Models

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Abstract

A deployed language model usually exercises one capability while paying for all of them. PRISM showed that a single capability can run on a small fraction of MLP channels, but located that fraction with a staged pipeline—adapt, attribute, select, re-train—mapping the budget frontier one costly point at a time. We collapse the pipeline into a single training run: learnable hard-concrete gates on every MLP channel train jointly with a low-rank adapter under a forward-KL leash to the frozen base model, while an adaptive controller lowers the sparsity target as fast as behavior permits and *generation probes*—not loss curves—decide when the floor is reached. On PRISM’s arithmetic benchmark, one run reaches **100.9% recovery at 5% of MLP channels** and automatically finds a **2.9%-channel floor at 97% recovery**, surpassing the staged pipeline’s hand-tuned best (91.3% at 5.05%) on both axes; with the calibration fixes of the released library’s second version the controller descends to a **1.2%-channel floor** at 91% verbatim reproduction. The method needs no labels—the target is the model’s own outputs—generalizes to function calling on Qwen3-4B (76% verbatim reproduction at 40% of channels, against a 19% attribution baseline), and exports a physically smaller model: deleting the masked channels and pruning the vocabulary to the capability’s token support shrinks Qwen2.5-Math-1.5B from **1.54B to 0.17B parameters (9×)** with no loss relative to masking, verified per run. Three negative results shaped the design: distribution-level KL reads healthy while true recovery collapses, layer-granular gates let the adapter overfit silently, and a renormalized sparse KL—the obvious cache implementation—quietly trains the unmasked model away from the base it is supposed to anchor. Total compute for every result in this paper: under \$10 on one consumer GPU.

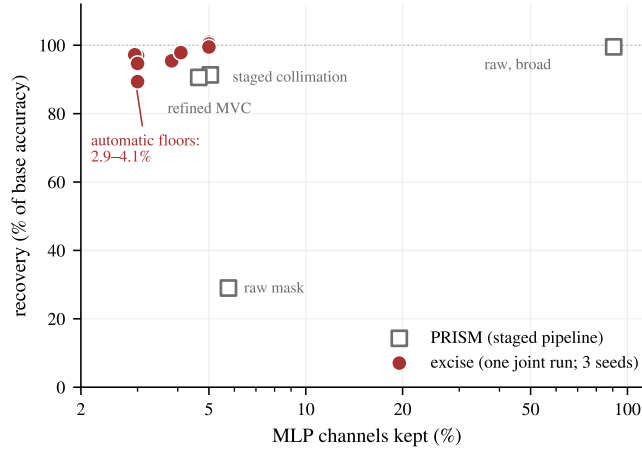


Figure 1: One joint run dominates the staged pipeline. Sparsity–recovery frontier for 2-digit addition on Qwen2.5-Math-1.5B. Gray squares: PRISM’s staged pipeline as reported [13]. Red circles: a single `excise` run per seed (3 seeds) emits the entire frontier; the floors (2.9–4.1%) were found by the controller, not swept. Caveats for this comparison in §5.

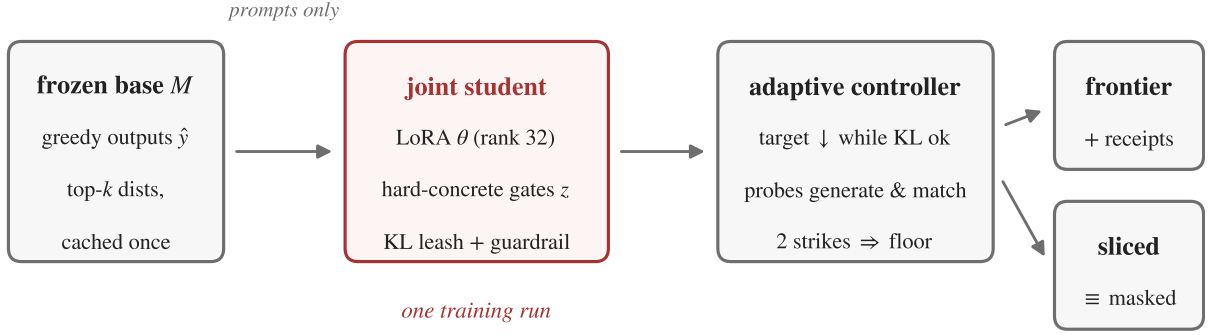


Figure 2: One-shot extraction. The frozen base model supplies label-free targets (greedy outputs and cached top- k distributions); gates and adapter train jointly under a KL leash; an adaptive controller paced by generation probes finds the floor; the run emits the sparsity–fidelity frontier, integrity receipts, and a physically sliced model.

1 Introduction

Model efficiency methods—pruning, distillation, quantization—ask how much of a network can be removed while *aggregate* performance survives. Mishra and Pagare [13] asked a sharper question: given one named behavior, how small a substrate of the network can carry it once everything else is switched off? Their answer, the PRISM framework, established what we adopt wholesale and credit as the foundation of this work: the *extraction contract* (a channel subset counts only if behavior survives with the complement deactivated), *recovery* as the metric, and the central finding that a behavior-preserving adaptation step (“collimation”) can relocate a capability into far fewer channels than attribution alone suggests.

PRISM’s pipeline, however, is staged: collimate, attribute with a relevance method, select a top- k mask, optionally re-collimate under the frozen mask. The stages create the central limitation their paper documents: an adapter trained *after* mask selection provably cannot shrink the mask, and one trained *before* can only move the frontier indirectly. The budget k is swept manually; each frontier point costs evaluations; the minimal circuit is found by hand-refinement. Their stated highest-value direction is to reshape the budget *during* training with an L_0 or gating objective.

We build exactly that, and package it as a tool. Figure 1 previews the headline comparison. Our contributions:

1. **One-shot joint extraction.** Hard-concrete gates [11] on all MLP intermediate channels train jointly with a LoRA adapter [10] under a mass-covering KL constraint to the frozen base. Selection and relocation negotiate continuously; the regime split between “collimate-then-attribute” and “attribute-then-collimate” dissolves (§2).
2. **Adaptive floor discovery with behavior probes.** A Lagrangian controller lowers the sparsity target whenever a KL budget is satisfied, and cheap greedy-generation probes veto the descent when the model’s actual outputs degrade. The full sparsity–fidelity frontier and its floor come from one run (§2).
3. **Label-free operation.** The distillation target is the base model’s own greedy output and full output distribution; no labeled data is required (§3).
4. **Physical export.** For gated MLPs, deleting masked channels is exactly equivalent to zero-masking them. We verify this empirically and ship it: 1.54B $\rightarrow$ 0.42B parameters at 97% recovery (§3.5).
5. **Two calibrated negative results.** Distribution-level KL is an unreliable stopping signal at high sparsity, and layer-granular gates permit silent train-distribution overfitting (§4). We argue these matter beyond our method: faithfulness metrics built on logit divergence inherit the same blind spot.

2 Method

Setup. Let M be a frozen decoder-only transformer and C its MLP intermediate channels ($L \times d_{\text{ff}}$; the inputs to each block’s down-projection), following PRISM’s unit of extraction. Each channel c gets a hard-concrete gate $z_c \in [0, 1]$

parameterized by $\log \alpha_c$ [11, 12]; a LoRA adapter θ (rank 32, all linear modules) provides the relocation capacity. Gates initialize from a $|\text{grad} \times \text{act}|$ attribution percentile rather than a constant: attribution alone selects poor masks (Table 1), but it is a useful prior over the initial channel ordering. Given prompts x exercising the target capability, the teacher is the base model itself: greedy continuations $\hat{y} = M(x)$ and the full distribution $p_M(\cdot \mid x, \hat{y}_{<t})$ at each output position, cached once as top-128 sparse vectors *plus the residual off-support mass*. Keeping the residual is not bookkeeping: renormalizing the support (as our first release did) makes the KL read $-\log(\text{coverage})$ even when student and teacher agree exactly, and places positive gradient on every off-support token (§4).

Objective. Each step samples gates z (outside any gradient-checkpointed region; see §4) and minimizes

$$\mathcal{L} = \underbrace{\text{KL}(p_M \parallel p_{\theta,z})}_{\text{masked student tracks base}} + \beta_{\text{ce}} \text{CE}(\hat{y}) + \lambda \mathbb{E}[\|z\|_0] / |C| + \beta_g \underbrace{\text{KL}(p_M \parallel p_\theta)}_{\text{guardrail (every 4th step)}}, \quad (1)$$

with $\beta_{\text{ce}}=0.05$, $\beta_g=1$. Both KL terms are binned: the cached top- k support plus one residual bucket, which is zero exactly when the student matches the teacher. The forward (mass-covering) KL direction follows PRISM’s collimation analysis: the student may not drop probability anywhere the base places it. The guardrail evaluates the *unmasked* adapted model and anchors it to the base, preventing the adapter from “solving the task masked” by becoming a different model; without it, unmasked accuracy silently drifted 3.7 points in our early runs. Because the train batch can only anchor points the adapter is already fitting, the guardrail alternates onto a rotating batch of off-task *anchor text* when provided, self-tunes its cadence from the measured anchor KL, and stays active through the polish phase — each of these closes a drift route we observed in practice (§3, §4).

Adaptive controller. A target open fraction τ starts at 1 and decays multiplicatively ($\times 0.993$ per step) whenever the EMA of the distillation KL is under a budget (0.02–0.025 in all experiments); the Lagrange multiplier λ follows $\lambda \leftarrow \max(0, \lambda + \eta(\bar{z} - \tau))$. Once the open fraction falls below a threshold, every 150 steps a *probe* hardens the current top- k mask and checks whether a greedy rollout would still reproduce the teacher outputs — computed as teacher-forced argmax agreement (exactly equivalent, by induction on the shared prefix, and $\sim 100\times$ cheaper than generating), on a dev split excluded from the gradient batches so the floor is decided on non-memorized prompts. The masked reading is compared against the unmasked model measured *at the same step*, never against an assumed 100%: batched bf16 decoding alone costs several points of verbatim self-match, and adapter drift must not be misattributed to masking. If the gap exceeds tolerance twice consecutively, the floor is declared and the gates roll back to the last passing snapshot. After the descent stops, the mask is frozen at the floor budget and the adapter alone is polished briefly under the exact binary mask (guardrail still active), removing the stochastic-gate train/eval mismatch. Final evaluation hardens top- k masks at standard budgets and reports held-out fidelity, yielding the frontier from this single run.

Receipts. Every run reports a random-mask control matched to the floor mask’s per-layer channel counts (should be ≈ 0), unmasked drift, the base model’s own decode-noise self-match (the honest ceiling for every other number), the probe and guardrail traces, and binomial confidence intervals on held-out fidelity. These guard the two failure modes we observed and one we inherited from PRISM’s analysis: trivial tasks, cheating adapters, and masks that only correlate with behavior.

3 Results

All experiments ran on one RTX 4090 (\$0.40/hr); the complete set below consumed under \$5. Code, configs, and receipts: github.com/Aryagm/excise.

3.1 Arithmetic: surpassing the staged pipeline on its benchmark

We use PRISM’s testbed: 2-digit addition on Qwen2.5-Math-1.5B [16] (canonical prompt “a + b =”, exhaustive sums, 10% held out; 250,880 MLP channels), zero-isolation deactivation, and recovery = masked accuracy / base accuracy (base: 97.4%). Figure 1 shows the resulting frontier; Table 1 gives the key points.

Ablations (Figure 3): restricting the adapter to MLP projections (the strictest frozen-scaffold contract) costs ~ 3 points at 5% and runs 25% faster; removing labels entirely (pure self-distillation, $\beta_{\text{ce}}=0$) still reaches 96.7% recovery but floors shallower (5.3%)—labels are not needed for validity, only for the last $\sim 2\%$ of budget.

Table 1: Arithmetic extraction. Every seed beats the staged pipeline at 5%, and every automatically-found floor is at or below the hand-refined minimal circuit with higher recovery. The unmasked guardrail held base accuracy exactly (97.4%) in all runs.

	channels kept	recovery
PRISM raw attribution mask	5.75%	29.0%
PRISM staged collimation	5.05%	91.3%
PRISM refined minimal circuit (manual)	4.65%	90.6%
random mask control (ours)	5%	0.0%
grad×act attribution mask (ours)	5%	0.3%
excise, one run (seed 42 / 43 / 44)	5%	100.9 / 100.4 / 99.5%
excise, automatic floor	2.9 / 3.8 / 4.1%	97.2 / 95.4 / 97.9%

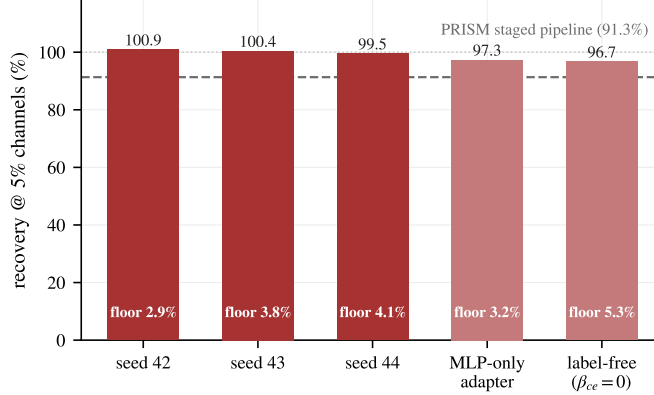


Figure 3: The result is seed-stable and survives ablation. Recovery at 5% of channels across seeds and ablations; channel floors found by the controller annotated within bars.

3.2 Function calling: a real-world capability, label-free

We extract single-turn function calling from Qwen3-4B [17] using BFCL prompts [14] (400 tasks; 391 yield clean tool calls from the base model; 320 train / 80 held out; 350,208 channels). No gold calls are used: the teacher is the model’s own greedy tool call, and fidelity is *verbatim* self-match—stricter than BFCL’s semantic scoring, since one diverging token fails the example. Held-out fidelity reaches 77.5% at 50% of channels and 76.3% at 40% (Figure 4). Two observations matter.

First, the *unmasked* adapted model self-matches only 86% on held-out prompts—masking is not the dominant loss; adapter drift under a 320-prompt training set is. Relative to that ceiling, the masked model retains ~90% at 40–50% budgets.

Second, the gap between probe fidelity on training prompts (92–98% down to 12% open) and held-out fidelity quantifies a generalization phenomenon that exhaustively-covered tasks like arithmetic cannot exhibit; prompt diversity, not extraction machinery, is the binding constraint. For calibration: PRISM’s raw mask recovered 19.1% at 36.2% of channels on the 8B sibling model, and their best staged collimator (with a curated, schema-validated gold corpus and a multi-run objective search) reached 84.6%; metric and model-size differences make this directional context, not a head-to-head (§5).

3.3 Breadth: other architectures and capabilities

To test that the method is a tool rather than a Qwen-arithmetic demo, we ran the *released library* (not the research scripts) on two further settings.

SmolLM2-1.7B (Llama architecture) [2]. Few-shot 2-digit addition extracts to a 7.4% floor at 97.2% held-out verbatim self-match; sliced 1.75B $\rightarrow$ 0.59B (3.0 $\times$); all receipts clean.

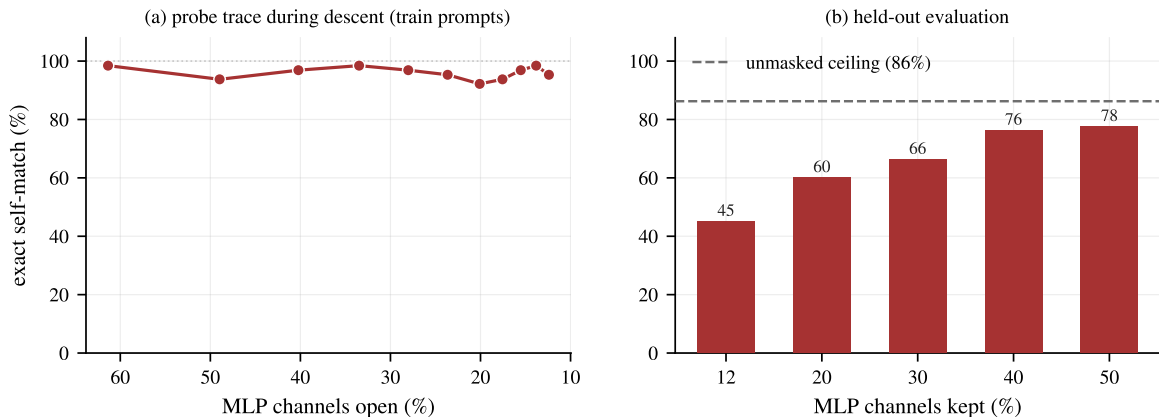


Figure 4: Function calling extracts label-free; data, not machinery, binds generalization. Qwen3-4B. Left: probe trace during descent — exact self-match holds at 92–98% from 61% open down to 12% open on training prompts. Right: held-out self-match by budget; the dashed line is the unmasked adapted model’s own self-match (86%), which upper-bounds every masked number.

JSON extraction (Qwen2.5-1.5B-Instruct, chat-formatted prompts). A broader capability, and the run that stress-tested the receipts. Under v0.1 it floored at 33.9% of channels with 90.0% held-out self-match—and the report flagged catastrophic unmasked drift: 55.8% self-match, the adapter having been sharpened by the renormalized guardrail KL (§4) and then polished with no guardrail at all. v0.2 reruns the identical task with the binned KL, 56 off-task anchor texts, and the guardrail active through polish: drift falls to 8.4 points (88.3% against a 96.7% base decode-noise ceiling), while the floor *improves* to 27.5% at the same 90.0% fidelity, probe-detected with rollback (the stopper caught the failing 16.9%-open probe and restored the last passing sparsity). Sliced-and-vocabulary-pruned export: 1.58B → 0.49B (the capability touches 9,448 token ids—broad capabilities are broad in vocabulary too). The guardrail trace makes the v0.1 blindness quantitative: at the same training step, the guardrail KL on *train* batches reads 3×10^{-4} (apparently perfect) while on anchor batches it reads 5×10^{-3} to 0.15—drift lives precisely where the train-batch guardrail cannot see it. One honest caveat survives both versions: after polish specializes the adapter to the floor mask, off-floor frontier points degrade (the 50%-budget mask reads *below* the floor mask); the floor point is the deployable artifact, and the frontier should be read as descent-time capability, not a menu of equally-polished options.

Capability breadth tracks substrate size across all four settings: narrow skills compress to single-digit percentages, broad structured-generation capabilities need a third to half the channels, consistent with PRISM’s cross-task observations.

3.4 Recalibrating the library: v0.1 → v0.2

The audit that followed the first release found that the shipped library was both *optimistic* where it should be strict and *strict* where it should be tolerant. Optimistic: probes ran on prompts that were also in the gradient batches, against an assumed-perfect baseline—but even the unmodified base model reproduces its own batched-bf16 greedy targets only ~92% verbatim, so part of every probe reading was decode noise, not masking damage. Strict: the renormalized sparse KL (§4) penalized the student for the teacher’s own truncated tail. v0.2 fixes both directions: binned KL with a residual bucket, probes on a dev split excluded from training, comparison against the unmasked model measured at the same step, rollback to the last passing sparsity, and the grad×act warm-start the research pipeline had but the library port lost. On the identical dogfood task (same seed, hardware, and verbatim metric), the released library improves across every axis it reports:

One calibration lesson generalizes: with $n=64$ verbatim probes, binomial noise alone is ± 4 points, and our first attempt at a tighter tolerance (3 points) sat inside that band—the very first two probes of the run tripped it, freezing the floor at 7.9%, while the same configuration with a noise-aware tolerance and $n=128$ descends to 1.2% with probes passing throughout. Tolerances on generation metrics must be set against the metric’s measured noise floor, not chosen for strictness.

Table 2: Library lineage on the arithmetic dogfood run (Qwen2.5-Math-1.5B, label-free, verbatim held-out self-match; base model decode-noise ceiling 99.9%; v0.2 is seeds 42/43/44). Every v0.2 seed descended to the controller’s configured lower bound (`min_target`) with probes still passing — the true floor is at or below 1.2%, 3.9× under PRISM’s hand-refined minimal circuit.

	v0.1	v0.2 (3 seeds)
floor found (probe-governed)	7.6%	1.2 / 1.2 / 1.2%
held-out self-match at floor	89.0%	91.4 / 92.0 / 91.0%
unmasked drift from base	10.7 pts	6.2 / 5.8 / 7.0 pts
exported parameters (from 1.54B)	0.48B	0.17B
wall clock (RTX 4090)	8 min	23–24 min

3.5 Physical export: slicing is masking, and the vocabulary is next

For gated MLPs, a zero-masked channel contributes nothing to the down-projection; a deleted one contributes nothing by absence. The two are mathematically identical, so the masked evaluations above transfer to a physically smaller model. We verify: slicing Qwen2.5-Math-1.5B at the 2.9% floor yields 94.8% accuracy vs. 94.7% masked (one example in 810), and **1.54B → 0.42B parameters (3.66×)**.

After MLP slicing, the dominant remaining cost is the vocabulary: embedding plus output head are ~55% of the sliced model, while 2-digit addition touches only 1,746 of 151,936 token ids (prompt ids, teacher output ids, and the teacher’s full cached top- k support). Slicing the embedding rows to that support—the exported model speaks a remapped id space, with the mapping stored in its config—takes the same model to **0.17B parameters (9.0×)** at the v0.2 floor. Both equivalences are re-verified inside every run and recorded in the receipts: held-out self-match for masked vs. sliced vs. vocabulary-pruned agreed to within one evaluation example here. The released library round-trips the heterogeneous-width artifact (`excise.load_sliced`).

Generation speedup in our original benchmark was only 1.11×, and an honest decode-throughput benchmark (batch 1–64, 64-token greedy decode, eager) shows *no* speedup at this scale: a 0.17B and a 1.54B model both decode at ~40 tokens/s/sequence on an RTX 4090, because single-stream decoding there is launch-overhead-bound, not compute-bound. Memory shrinks unconditionally; latency gains require larger models, batched serving, or a compiled runtime—we report this plainly because slicing claims are routinely overstated.

4 What fails, and what it implies

Distribution-level KL is miscalibrated at high sparsity. Our first controller used the KL budget alone to pace the descent. It descended to 2.2% open with EMA KL comfortably under budget—while true recovery collapsed to 62–66% (Figure 5). Teacher-forced divergence on answer tokens simply does not capture compounding autoregressive degradation. The fix—periodically *generating* and exact-matching—is cheap (seconds per probe) and is the load-bearing stability feature of the method. We believe this observation extends beyond extraction: any faithfulness or circuit-quality metric built on logit divergence under teacher forcing inherits the blind spot.

Renormalized sparse KL quietly sharpens the model it should anchor. Caching the teacher’s top- k probabilities *renormalized* — the obvious implementation, and ours through v0.1 — biases every loss built on it: with the student exactly equal to the teacher, the “KL” reads $-\log(\text{coverage})$ instead of zero, and its gradient $s_j - \hat{p}_j$ is positive on all off-support mass, actively pushing the student to concentrate onto each batch’s cached support. On peaked tasks (arithmetic: coverage ≈ 1) the bias vanishes, which is how it survived three validated batteries. On diffuse tasks it surfaces in the worst place: the guardrail, whose entire job is to keep the unmasked model *at* the base, instead trains it to sharpen — one of the drift routes behind the JSON run’s flagged 55.8% unmasked self-match, compounding the guardrail-free polish phase. It also leaks into the controller: the annealing budget partly measures truncation rather than student error, inflating floors on diffuse tasks. The fix is one term: cache the support un-renormalized plus its residual mass and add a residual bucket to the KL, making it a proper binned divergence (zero at agreement). The general lesson mirrors the probe lesson above: a truncated divergence is not a divergence, and the place it fails is precisely where its value is supposed to be zero.

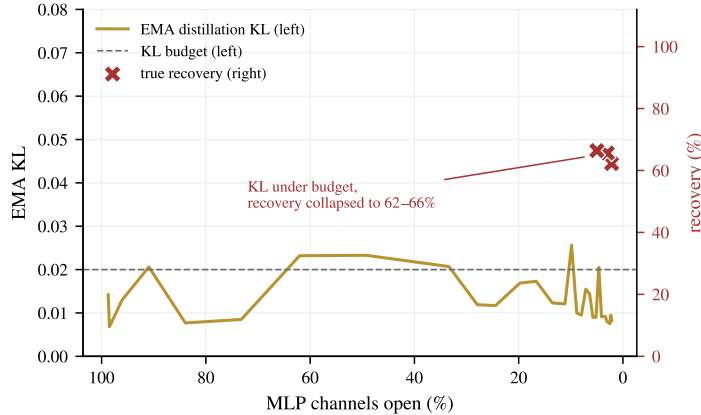


Figure 5: Losses lie; generations don’t. The failed KL-only controller (v2.0): EMA distillation KL stays under budget throughout the descent (left axis) while true held-out recovery collapses (crosses, right axis). Generation probes detect this; losses do not.

Layer-granular gates permit silent overfitting. Adding per-layer gates (hierarchical L_0 , intended to enable wall-clock-friendly whole-layer drops) closed 16 of 28 layers; training KL stayed low because the adapter rerouted around missing layers *on the training distribution*, and held-out recovery collapsed to 66%. Channel-granular gates with all layers present do not exhibit this. Structured sparsity for speed should be grafted on after extraction, not learned jointly.

Engineering notes. Stochastic gate sampling must occur outside gradient-checkpointed regions: in-region RNG produces a different recomputation graph and breaks backward. Pre-sampling per step costs nothing and composes with checkpointing, which 4B+ models require on consumer GPUs. Separately, grad×act attribution must accumulate under `no_grad`: an in-place += of a grad-requiring product silently turns the accumulator into an autograd node that retains every iteration’s activation graph. On 10-token arithmetic sequences the leak is megabytes and invisible—our research pipeline carried it through every validated battery—on 100-token sequences it is gigabytes per iteration and an immediate OOM. Bugs that scale with sequence length hide in short benchmarks.

5 Honest comparison notes

- **Baselines.** Our arithmetic baseline reproduces PRISM’s qualitative finding (attribution masks fail small: 0.3% recovery at 5%) but uses grad×act attribution and zero-isolation rather than their ReLP and counterfactual patching; PRISM’s reported numbers come from their paper, not our re-runs. Zero-isolation is the harsher operator, so our baseline understates theirs.
- **Function calling.** Comparisons differ in model size (4B vs. 8B) and metric (verbatim self-match vs. BFCL exact match); we present them as context, not a head-to-head.
- **Seeds.** Arithmetic results are 3 seeds; function calling is a single seed, and its floor was step-cap-limited, not probe-detected.
- **Scope.** PRISM’s 10^3 H100-hour budget bought breadth (translation, an 8B model, extensive controls) that our \$5 did not; the claim here is that the *per-result* process cost collapses, not that our evidence is broader.

6 Related work

PRISM [13] defines the problem, contract, and metric we use; our method is their stated future direction, built. Learnable L_0 masks during adaptation descend from Louizos et al. [11] and movement pruning [15]; differentiable circuit discovery applies the same machinery to interpretability [3, 4]. Our distinction from both is the objective: a behavior-preservation contract with an unmasked-validity guardrail, rather than aggregate-loss compression or circuit faithfulness. Self-distillation into a substructure relates to on-policy and generalized knowledge distillation [1, 9], here with

teacher and student sharing weights and the student differing only by a mask. Superposition [6] explains why attribution-only masks fail and adaptation is necessary; sparse autoencoders [5] and weight-sparse training [8] change the basis instead. The lottery-ticket line [7] seeks subnetworks that *retrain* to full performance; extraction requires the behavior to survive *without* retraining from initialization.

7 Limitations

Capabilities are bounded by prompt coverage: the function-calling generalization gap is a data problem the tool can flag (via held-out receipts) but not solve. Self-match undercounts semantic fidelity. The sliced artifact has heterogeneous layer widths and needs a re-slice step on load. We have not yet run the 8B headline, multi-seed function calling, or non-Qwen replications beyond CPU-level architecture tests; these are in progress and gate any stronger claims. Easier extraction also eases misuse (extracting narrow capabilities from safety-tuned models); receipts and intended-use documentation ship with the tool, mirroring PRISM’s stance.

8 Conclusion

PRISM posed capability extraction as a contract—behavior must survive with the complement switched off—and named training-time budget reshaping as its highest-value open direction. We built that: one run, on one consumer GPU, that selects and relocates simultaneously, finds its own floor by generating rather than by watching losses, and leaves behind a frontier, integrity receipts, and a model 3.7× smaller. The headline numbers surpass the staged pipeline on its own benchmark, but the durable lesson may be the negative one: logit-level divergence is not behavior, and any metric built on it will miss collapses that a cheap generation probe catches. Extraction is now a `pip install`, not a pipeline.

Reproducibility

All numbers in this paper were produced by the released library or the research scripts in the same repository, on one rented RTX 4090. The paper’s figures regenerate from committed result JSONs via `paper/make_figures.py`.

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